PROBABILITY AND STATISTICS

Directions for questions 1 and 2: Select the correct alternative from the given choices.

If the arithmetic mean of two numbers is 45 and their geometric mean is 36, the two numbers are _______.

a) 45, 45

b) 18, 72 (Correct Answer - ✅)

c) 6, 216

d) 30, 60

Solution:

Let a and b be the two numbers.

Given $\frac{a+b}{2} = 45;\ \sqrt{ab} = 36$

$a + b = 90;\ ab = 1296 = 16(81) = 18(72)$

Solving the above equations we will get

$\{a, b\} = \{18, 72\}$

Choice (B)

Directions for questions 1 and 2: Select the correct alternative from the given choices.

The mode and mean of a moderately symmetric distribution are 17 and 8 respectively. The median of the distribution is _______.

a) 10

b) 9

c) 11

d) 17

Solution:

We know that

Mode = 3 median – 2 mean

$$\text{Median} = \frac{\text{mode} + 2\text{mean}}{3}$$

$$\frac{17 + 2 \times 8}{33} = \frac{33}{3} = 11.$$

Choice (C)

Directions for questions 3 and 4: Type in your answer in the input box provided in the question.

The standard deviation of $4x_i + 3$, where i = 1, 2, 3. . . is 12. The variance of $3x_i$ is

Solution:

Given the standard deviation of $4x_i + 3$ or of $4x_i$ is 12,

The standard deviation of x_i is 3

The standard deviation of $3x_i$ is 9

Hence variance of $3x_i$ is 81.

Ans : (81)

Directions for questions 3 and 4: Type in your answer in the input box provided in the question.

If the arithmetic mean of cubes of first 'n' natural numbers is 162, then n =

Solution:

The arithmetic mean of cubes of n natural numbers is $\frac{n(n+1)^2}{4}$. $\left(\because \frac{(n(n+1)/2)^2}{n}\right)$

Given $\frac{n(n+1)^2}{4} = 162$, n (n + 1)^2 = 8(81)

n = 8 satisfies the above equation.

Ans : (8)

> ***Directions for question 5:*** Select the correct alternative from the given choices.

The range of 3, 5, 4, 8, 12, x, 15, 17 is 16. x can be ______.

- a) 2
- b) 18
- c) 19
- d) 4

Solution:

Let x be the minimum value. Then range = maximum value – minimum value.
$16 = 17 - x \Rightarrow x = 1$
If x is the maximum value then
$16 = x - 3 \Rightarrow x = 19$.
∴ x can be 1 or 19. Choice (C)

> ***Directions for questions 6 and 7:*** Type in your answer in the input box provided in the question.

The median of the first 15 prime numbers is

Step 1: List the first 15 primes

$2,3,5,7,11,13,17,19,23,29,31,37,41,43,47$

Step 2: Recall median rule

- Median = middle value when numbers are arranged in order.
- For **odd count (n = 15)**, median is the $n+12$-th term.

$15+12=8$

Step 3: Identify the 8th prime

The 8th prime in the list is **19**.

☑ Final Answer:

The median of the first 15 prime numbers is **19**.

> ***Directions for questions 6 and 7***: Type in your answer in the input box provided in the question.

The mean deviation of the distribution 45, 55 is

Solution:

The mean deviation of a and b is $\frac{|a-b|}{2}$.

$\therefore$ The mean deviation of 45 and 55 is $\frac{|45-55|}{2} = 5$. Ans : (5)

> ***Directions for questions 8 to 13***: Select the correct alternative from the given choices.

When three dice are rolled, what is the probability that all the three dice show the same number?

- a) $\frac{1}{6}$
- b) $\frac{1}{36}$
- c) $\frac{1}{216}$
- d) $\frac{1}{108}$

Solution:

When three dice are rolled, the total number of outcomes is 216. The favourable outcomes are (1, 1, 1) (2, 2, 2) (3, 3, 3) (4, 4, 4) (5, 5, 5) (6, 6, 6) i.e. 6 in number.

$\therefore$ Required probability = $\frac{6}{216} = \frac{1}{36}$. Choice (B)

Directions for questions 8 to 13: Select the correct alternative from the given choices.

If five coins are tossed, the probability of getting 4 tails and one head is _______.

- a) $\frac{27}{32}$
- b) $\frac{3}{16}$
- c) $\frac{5}{32}$
- d) $\frac{5}{16}$

Solution:

We know that, when n coins are tossed, the probability of getting r tails is $\frac{{}^{n}C_{r}}{2^{n}}$.

Required probability is = $\frac{{}^{5}C_{4}}{2^{5}}$ = 5/32.

Choice (C)

Directions for questions 8 to 13: Select the correct alternative from the given choices.

A bag contains 5 white balls, 4 green balls and 3 red balls. If two balls are selected from the bag, the probability that the two balls are white is _______.

- a) $\frac{5}{33}$
- b) $\frac{2}{11}$
- c) $\frac{1}{22}$
- d) $\frac{1}{11}$

Solution:

Total number of balls is 12. Two balls can be selected from 12 balls in ${}^{12}C_{2}$ = 66 ways. Two white balls can be selected from 5 white balls in ${}^{5}C_{2}$ = 10 ways required probability = $\frac{10}{66} = \frac{5}{33}$.

Choice (A)

Directions for questions 8 to 13: Select the correct alternative from the given choices.

Three distinct natural numbers are selected from the first 10 natural numbers. The probability that the product of the numbers is odd is ______.

- a) $\frac{5}{6}$
- b) $\frac{1}{6}$
- c) $\frac{11}{12}$
- d) $\frac{1}{12}$

Solution:

Three natural numbers can be selected from 10 natural numbers in ${}^{10}C_3$, i.e., = 120 ways.
When three numbers are odd, then the product of the numbers is odd. Number of favourable outcomes = ${}^{5}C_3 = 10$

required probability = $\frac{10}{120} = \frac{1}{12}$. Choice (D)

Directions for questions 8 to 13: Select the correct alternative from the given choices.

Two cards are selected from a pack of cards. What is the probability that at least one card is a king?

- a) $\frac{188}{221}$
- b) $\frac{33}{221}$
- c) $\frac{15}{221}$
- d) $\frac{106}{221}$

Solution:

Number of ways in which two cards can be selected from 52 cards ${}^{52}C_2 = 26\ (51)$.

Let E be the event that atleast one card is a king.

$\bar{E}$ is the event that no card is a king.

$n(\bar{E}) = {}^{48}C_2 = 24\ (47)$.

$P(\bar{E}) = \frac{24(47)}{26\ (51)} = \frac{188}{221}$

$P(E) = 1 - P(\bar{E}) = 1 - \frac{188}{221} = \frac{33}{221}$. Choice (B)

***Directions for questions 8 to 13*:** Select the correct alternative from the given choices.

A and B are independent events such that P(A) = 3/4 P(B) =2/3. P(AUB) = ______.

- a) $\frac{1}{12}$
- b) 1
- c) $\frac{11}{12}$
- d) $\frac{1}{2}$ (Your answer - ✕)

***Directions for question 14*:** Type in your answer in the input box provided below the question.

The probability of a random variable X is as follows. What is the expected value of X?

$X = x_i$	1	2	3	0
$P(x = x_i)$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$	$\frac{1}{8}$

Solution:

$$E(x) = \sum x_i P(x = x_i)$$

$$= 1\left(\frac{3}{8}\right) + 2\left(\frac{3}{8}\right) + 3\left(\frac{1}{8}\right) + 0\left(\frac{1}{8}\right)$$

$$= \frac{3+6+3+0}{8} = \frac{12}{8} = \frac{3}{2}.$$

Ans : (1.5)

Directions for question 15: Select the correct alternative from the given choices.

If A and B are two mutually exclusive and collectively exhaustive events such that P(A) = 5/8, then $P(\overline{B})$ = ______.

- a) $\frac{5}{8}$
- b) $\frac{3}{8}$
- c) $\frac{1}{2}$
- d) $\frac{3}{4}$

Solution:

A and B are two mutually exclusive events,

$\therefore P(A) + P(B) \leq 1.$ ...(1)

A and B are collectively exhaustive.

$\therefore P(A) + P(B) \geq 1$ ....(2)

$(1), (2) \Rightarrow P(A) + P(B) = 1.$

$P(A) = 1 - P(B) = P(\overline{B})$

$\therefore P(\overline{B}) = P(A) = \frac{5}{8}.$ Choice (A)